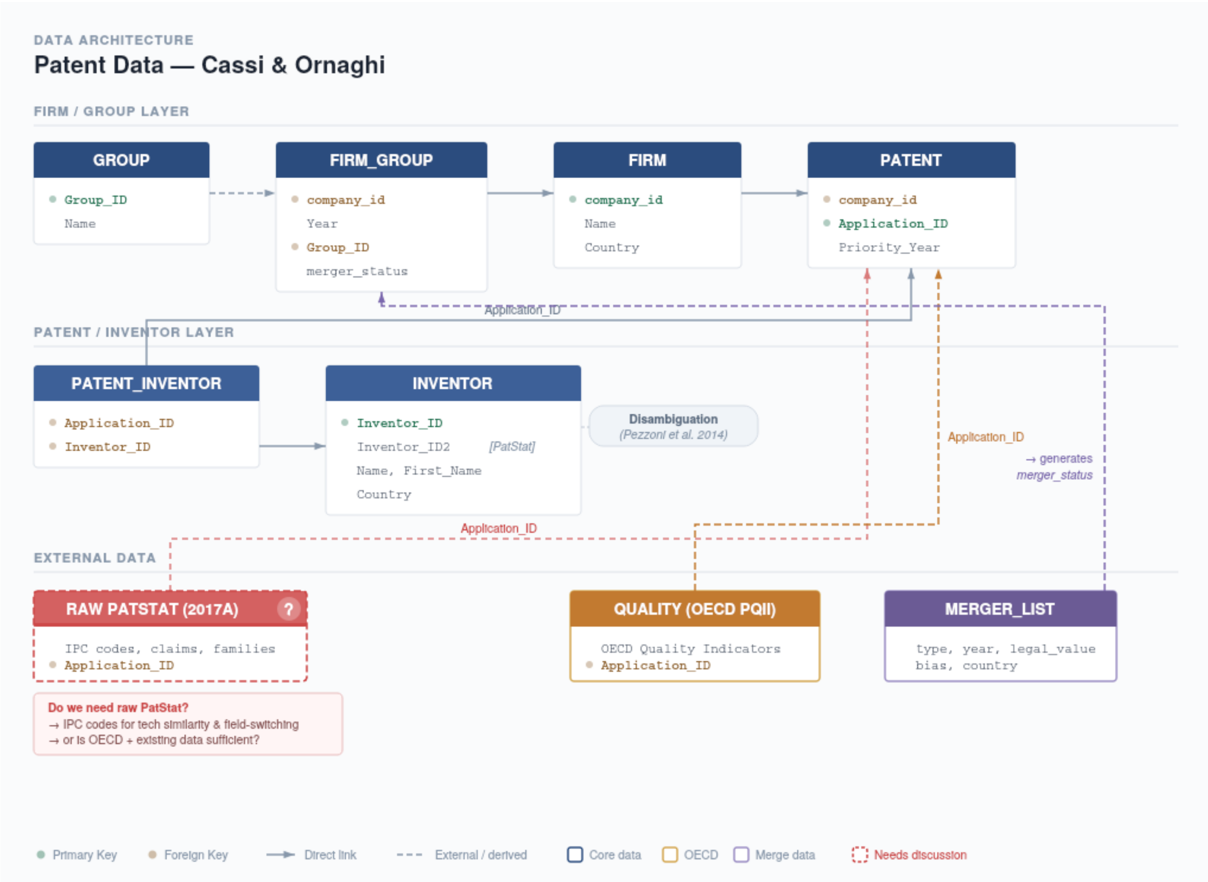
Data Discussion

After the Acquisition – Stayers' Innovation in Pharma M&A

Replication Package Overview

The diagram below summarizes the core data structure from the replication packages and additional sources.



Additional Data

- **Quality:** The OECD Patent Quality Indicators (PQII) seem like the most straightforward option. They offer composite quality scores, forward citation counts, and sub-aggregation indicators, all linkable via `Appln_ID`. I've already obtained the OECD data and also raw citation counts separately as a robustness check.

- **Direction (field-switching):** I need IPC codes to see whether stayers shift technological fields after the acquisition. The `T_inventors_tech_similarity` file from the replication package provides some similarity measures. I'm not sure yet whether I should construct inventor-level IPC vectors from scratch.

This raises the key open question: **do I need the raw PatStat data?** Access to this would provide IPC codes and patent families directly, enabling me to estimate technological similarity using IPC cosine similarity (Verginer et al., 2025).

Points to discuss

1. **Tech similarity:** The existing `T_inventors_tech_similarity` file uses group-level patent aggregation. For my field-switching measure, should I stick with this or build inventor-level IPC vectors instead (cosine similarity as in Verginer et al. 2025)?

2. **PatStat access:** Necessary? How to get access?

3. **Financial data:** Your original paper used financial data, but Verginer et al.'s two-stage matching does not require it. Can we skip the additional financial controls, as these are proprietary data?

Next steps

1. Match and merge the data into a single database

2. Clarify matching strategy and outcome equations

3. Begin building the DiD approach